

MedReview (Hospital, Doctors and Service Review System)

Project Proposal Report

1. Introduction

When navigating today's healthcare system, patients face a significant challenge: finding reliable, objective information on the quality of care and administrative efficiency of medical providers. While online review platforms are common for hospitality and retail, healthcare reviews suffer from a fundamental trust deficit. Traditional platforms rarely use structured medical metrics, are vulnerable to rating manipulation, and do not verify if a reviewer actually received care from the provider they are rating.

MedReview is a secure, web-based healthcare transparency platform designed to close this trust gap. By introducing a multi-dimensional rating system backed by physical and digital evidence (such as medical bills or appointment confirmations), MedTransparency provides patients with a reliable, verified directory of doctors, hospitals, and specialized medical services.

2. Problem Statement

Selecting a healthcare provider is currently an opaque and stressful process for patients. Without reliable, structured data, patients and clinics face several critical issues:

- **Unverified and Manipulated Ratings:** Standard public review sites allow anyone to post anonymous, unverified feedback. This makes it easy for bad actors to post fake positive reviews to artificially boost a clinic's reputation, or post malicious, fake negative reviews to damage a qualified doctor's standing.
- **Overly Simplified Quality Metrics:** A simple five-star rating scale does not provide enough detail for healthcare decisions. Patients need to understand

specific aspects of their visit, such as the clinic's punctuality, the administrative staff's professionalism, and the physician's bedside manner and medical knowledge.

- **Opaque and Surprise Billing:** Medical billing is notoriously complex and unpredictable. Patients rarely know what a facility's billing practices are like before receiving care, which often leads to unexpected costs and financial stress.
- **Outdated Provider Affiliations:** Doctors frequently practice across multiple clinics, medical groups, and hospital systems. Static medical directories fail to keep pace with these shifting affiliations, leaving patients confused about where a specific doctor is currently admitting patients or performing procedures.
- **HIPAA and Misinformation Risks:** Public health forums often struggle with accidental leaks of Protected Health Information (PHI) or the spread of unverified medical claims. Traditional platforms lack structured moderation workflows to identify and mitigate these compliance risks quickly.

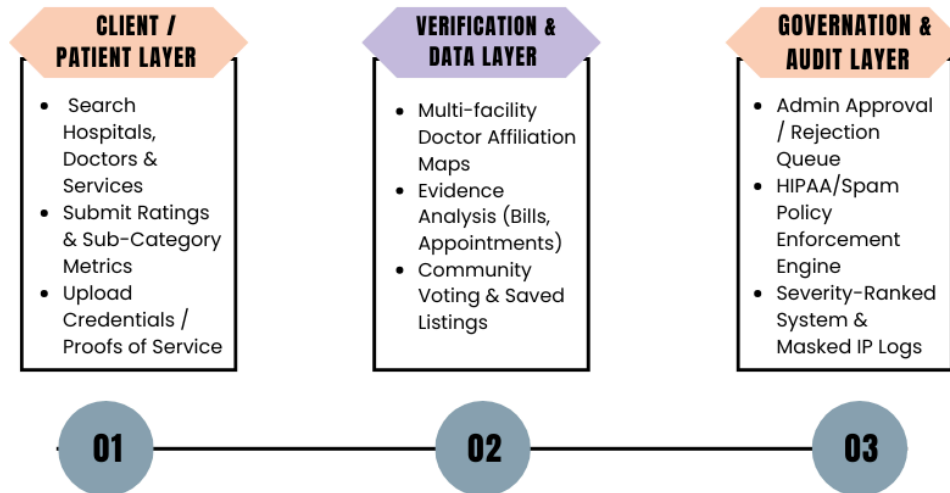
To address these challenges, we need a secure, relational platform that tracks medical entities, verifies patient reviews using concrete evidence, manages a clear moderation lifecycle, and maintains a secure system log for administrative auditing.

3. What Is My Solution?

MedReview solves these problems by providing a secure, web-based platform built on a relational database of 12 interconnected tables. It transforms online medical reviews from anonymous opinions into verified, trustworthy feedback.

The system is organized into three operational layers:

OPERATIONAL LAYERS



3.1 Patient & Provider Layer (Registry & Directory)

This layer serves as the single source of truth for the platform's medical directory. It organizes healthcare targets into three distinct reviewable categories: **Hospitals**, **Doctors** (categorized by their **Specialties**), and specific **Medical Services** (such as emergency rooms or pediatric departments) hosted at particular hospitals.

Because doctors often work at multiple locations, we use a connector table (**DoctorAffiliation**) to map these multi-facility relationships and clearly indicate where a provider spends most of their practicing time.

3.2 Engagement & Verification Layer (Reviews & Proofs)

When writing a review, patients rate their experience across four key areas: **Staff Friendliness**, **Punctuality**, **Medical Knowledge**, and **Bedside Manner**. These

sub-ratings are averaged to calculate the review's overall rating.

To ensure every review is genuine, patients submit a **ReviewProof** (such as an appointment code, a copy of a medical bill, or a document upload). Patients can also bookmark their favorite resources using **SavedListings** and vote on the helpfulness of other reviews using **ReviewVotes**.

3.3 Governance, Moderation & Audit Layer

To prevent spam and protect patient privacy, all new reviews are placed in a **PENDING** queue. Platform administrators review the uploaded proof, approve or reject the post, and add internal notes to document their decision.

To help maintain platform safety, users can submit a **ReviewReport** if they spot spam, offensive language, false medical information, or potential HIPAA violations. The system automatically records all administrative decisions, errors, and logins in the **SystemLog** table, masking IP addresses to protect user privacy.

4. Expected Outcomes

Once fully deployed, MedTransparency will deliver several key outcomes:

- **High-Trust Patient Reviews:** Reviews that successfully pass administrative verification receive a prominent "Verified Review" badge, removing the guesswork around which ratings to trust.
- **Targeted Quality Filters:** Patients can filter search results by specific operational traits, making it simple to find highly punctual clinics or doctors with excellent bedside manner scores.
- **Up-to-Date Provider Tracking:** Dynamic mapping of doctor-hospital relationships ensures patients always know exactly where a physician is currently practicing.
- **A Safe and Compliant Platform:** Community flag tools and admin moderation workflows allow the platform to quickly isolate and remove HIPAA violations or dangerous medical misinformation.
- **Audit-Ready Activity Logging:** Security teams can use detailed system logs to monitor admin decisions and detect potential system vulnerabilities while fully preserving patient confidentiality.

5. Unique Features vs. Competitors

Unique Feature	Why It Stands Out
Proof-Backed Review Engine	Standard review sites allow anyone to write anything. MedTransparency uses physical proof (bills, receipts, appointment codes) to verify that an actual visit occurred.
Granular Quality Scorecards	Instead of an oversimplified five-star scale, our reviews break down the patient experience into four key administrative and clinical performance areas.
Dynamic Doctor Mapping	We track doctors practicing across multiple facilities, showing patients exactly where their primary office is located.
HIPAA-Focused Reporting	Our reporting system categorizes flags by specific regulatory risks, prioritizing patient privacy protection and the removal of false medical claims.
Compliance-First Auditing	A secure, append-only system log records administrative actions and platform security events while automatically masking IP addresses to protect user identity.

6. Market Value & Business Opportunities

As healthcare costs continue to rise, patients are actively looking for high-quality, cost-effective care. MedTransparency addresses this demand by offering transparent, verified quality metrics.

Market Opportunities

- **Employer Health Benefits:** Businesses can offer MedTransparency to employees to help them locate high-quality, verified local care, lowering overall corporate insurance premiums.
- **High-Performing Practices:** Doctors and clinics with exceptional punctuality

and bedside-manner ratings can display their verified badges to stand out and attract new patients.

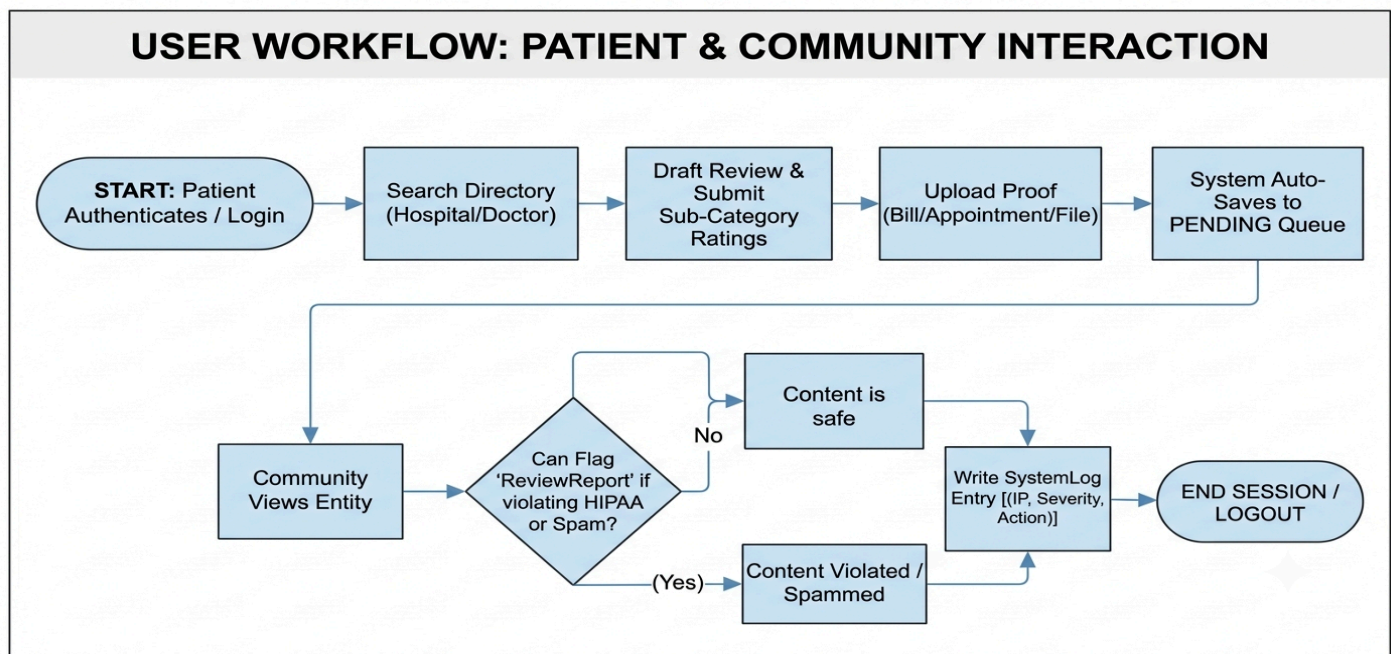
- **Healthcare Operations Planning:** Aggregated, anonymized sub-category rating data provides hospital administrators with valuable, localized feedback on clinical bottlenecks and service quality.
- **Hospital Network Management:** Large health networks can use our administrative dashboard to internally monitor patient satisfaction across all departments and affiliated physicians.

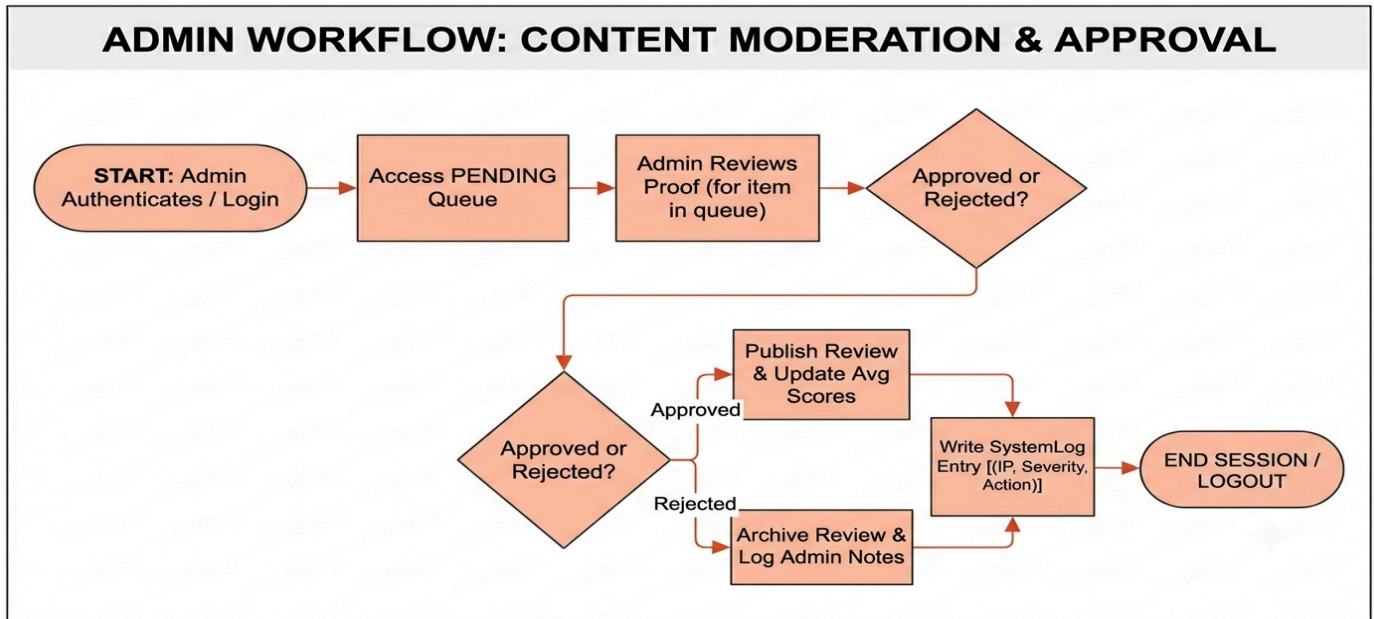
Business Value

- **Reduces search friction** for patients, making it easier to select the right care.
- **Highlights precise areas of improvement** for clinics (such as staff professionalism vs. wait times).
- **Protects patient privacy** and minimizes platform liability by tracking regulatory compliance.

7. Project Operational Flow

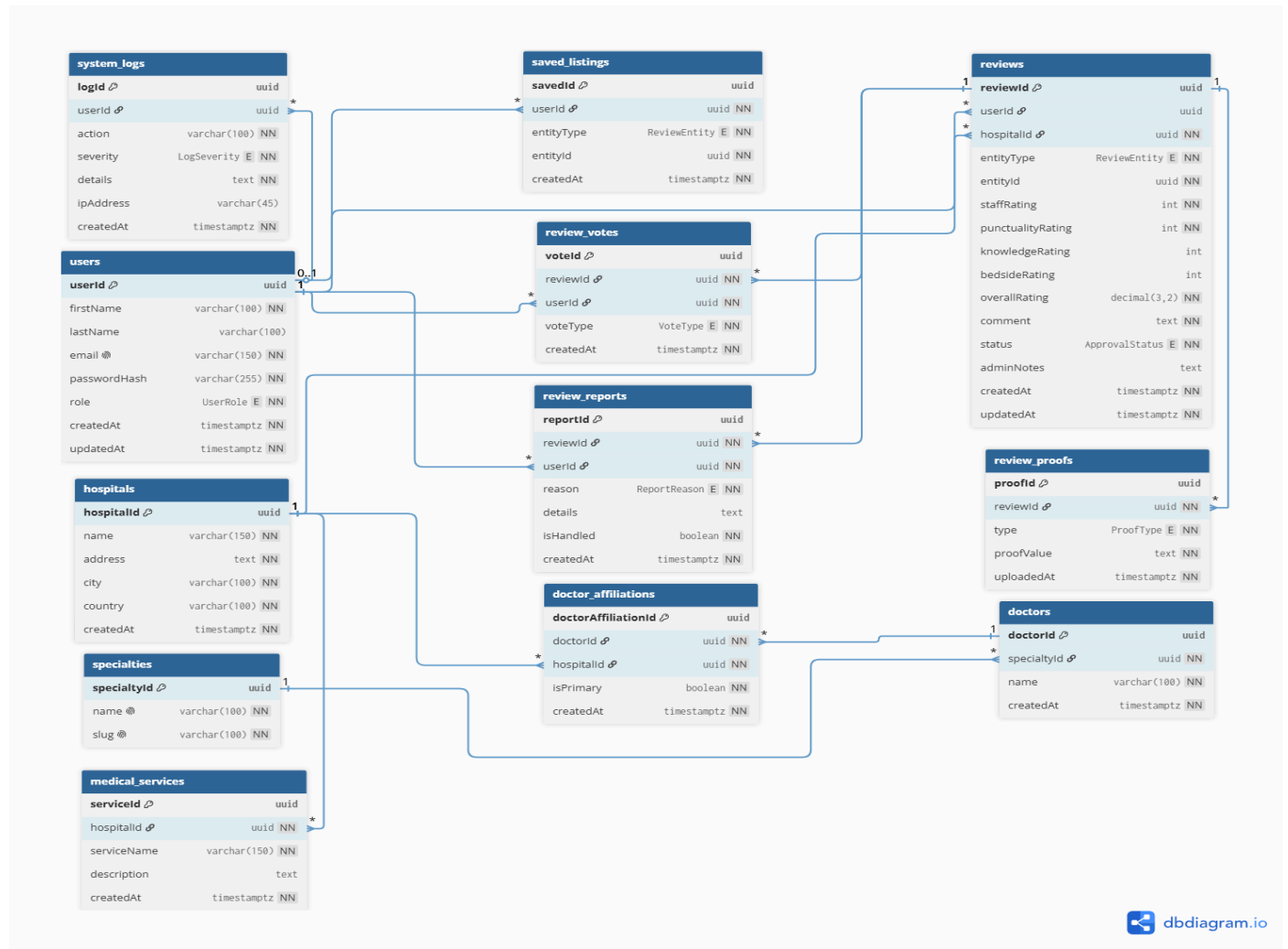
The diagram below illustrates how a review moves through the platform, from creation and verification to final publication and security logging:





8. Database Architecture (Entity-Relationship Diagram)

The database schema is designed to enforce data integrity and prevent inconsistencies, enabling fast queries across doctors, facilities, and patient feedback.



9. Features

Feature	Description	Tables We Use
Identity & Access Management	Handles secure user registration, logins, hashed passwords, and system role enforcement.	User
Comprehensive Clinical Registry	Maintains core directories for hospitals, clinics, practicing doctors, and specialties.	Hospital, Doctor, Specialty, MedicalService

Provider Affiliation Engine	Maps doctors practicing across multiple facilities, clearly indicating their primary workspace.	DoctorAffiliation
Polymorphic Review System	Allows users to submit structured, detailed ratings targeting a doctor, hospital, or specific clinical department using a unified schema.	Review
Physical Evidence Verification	Connects reviews to uploaded documents, billing data, or appointment numbers to confirm actual clinic visits.	ReviewProof
Community Curation Controls	Enables users to upvote helpful patient feedback and report spam, offensive language, or privacy issues.	ReviewVote, ReviewReport
Personalized Care Dashboards	Lets patients save specific medical entities to their personal dashboards for quick reference.	SavedListing
Security Auditing Log	Automatically records system actions, error states, and admin decisions with masked IP footprints.	SystemLog

10. Platform Functions by Role

Admin

- **Manage Users:** Onboard new administrators, audit platform activity, and suspend accounts violating guidelines.
- **Verify Evidence:** Review pending submissions in the proof queue, confirming uploaded documents or appointment numbers.
- **Moderate Content:** Move pending reviews to approved or rejected states, and

provide explanatory notes.

- **Enforce Compliance:** Investigate reported reviews for potential policy, spam, or privacy (HIPAA) violations.
- **Audit System Health:** Monitor severity-ranked system logs to identify potential operational errors or security threats.

User (Patient Client)

- **Explore Directory:** Search for local hospitals, doctors, and specific departments by location and specialty.
- **Share Experiences:** Write reviews evaluating staff friendliness, punctuality, clinical knowledge, and bedside manner.
- **Verify Visits:** Upload appointment codes or medical bills to earn a verified status badge on reviews.
- **Collaborate:** Upvote helpful reviews and flag posts containing spam, misinformation, or privacy leaks.
- **Bookmark Care:** Keep a personalized list of favorite doctors, clinical departments, and hospitals.

System (Automated Core)

- **Calculate Average Ratings:** Automatically update the average overall score of a medical entity whenever an admin approves a review.
- **Match Verification Codes:** Check submitted appointment hashes against verification databases to streamline verification.
- **Flag Privacy Infractions:** Automatically route reviews with multiple community reports directly to administrative holding for review.
- **Maintain Audits:** Record administrative decisions, database adjustments, system errors, and logins with privacy-masked IP footprints.

11. Relational Database Directory

The MedTransparency platform operates on 12 highly integrated tables. This directory explains each table's purpose and its active connections within the system.

1. **User:** Stores core profile and authentication details, including hashed credentials and platform roles (**USER** or **ADMIN**).
2. **Hospital:** Represents physical medical complexes, storing addresses, cities, and regional locations.

3. **Doctor:** Manages clinical practitioners, linking each doctor to their primary clinical specialty.
4. **DoctorAffiliation:** The connector table mapping doctors practicing across multiple hospital facilities, tracking their primary workspace.
5. **MedicalService:** Represents specific internal departments (such as cardiology, physical therapy, or emergency services) hosted at particular hospitals.
6. **Specialty:** Houses standard, board-recognized specialties (e.g., pediatrics, neurology) with URL-friendly slugs for easy searching.
7. **Review:** Captures patient feedback, including comments, admin notes, status states (**PENDING**, **APPROVED**, **REJECTED**), and five distinct quality scores. Since the targeted target can vary, we track this polymorphically using optional relationship keys (**hospitalId**, etc.) and a type enum.
8. **ReviewProof:** Stores proof metadata (appointment codes, bill summaries, or upload paths) to verify the legitimacy of a patient's visit.
9. **ReviewVote:** Tracks helpfulness votes cast on reviews, using composite database rules to prevent single-user vote duplication.
10. **ReviewReport:** Logs flagged reviews, detailing the reason for reporting (e.g., **HIPAA_VIOLATION**, **FALSE_INFORMATION**) and whether an admin has resolved it.
11. **SavedListing:** A junction table that acts as a bookmark folder, letting users save doctors, departments, or facilities to their personal dashboards.
12. **SystemLog:** An append-only log ledger for security. Records admin interventions, system warnings, and client activities, masking IP addresses to ensure user privacy.

12. Key Database Relationships

The table below outlines how the tables interact to maintain a consistent and reliable flow of data:

From Table	Relationship	To Table(s)	Description
User	1-M	Review, SystemLog, ReviewVote, ReviewReport, SavedListing	A user creates multiple reviews, votes, reports, bookmarks, and logs.

Hospital	1-M	MedicalService, Review, DoctorAffiliation	A hospital hosts various services, connects to practicing doctors, and receives reviews.
Doctor	1-M	DoctorAffiliation	A doctor can practice and hold clinical admittance rights across multiple hospital locations.
Specialty	1-M	Doctor	A single specialty classifies many practicing clinical professionals.
Review	1-M	ReviewProof, ReviewVote, ReviewReport	A review supports multiple verification proofs, votes, and compliance flags.
User & Review	M-M	ReviewVote (via Junction)	Many users can vote on many reviews, but database rules limit them to one vote per review to prevent manipulation.
User & Entity	M-M	SavedListing (via Junction)	Many users can bookmark many medical entities to build their personal favorites list.

13. Conclusion

The **MedReview** platform directly addresses the trust gap in online medical feedback. By transitioning from anonymous comment boards to a secure, relational database structure backed by verified proofs, MedTransparency equips patients with high-quality, verified data they can count on.

By evaluating clinical experiences across specific categories—staff friendliness, punctuality, physician knowledge, and bedside manner—alongside verifiable evidence, the platform ensures complete fairness, compliance safety (including active HIPAA

protections), and administrative transparency. This comprehensive plan establishes MedTransparency as a highly reliable resource for patients, providers, and corporate health groups alike.